1.0 INTRODUCTION

1.1 BACKGROUND OF CASE STUDY

Indigenous languages represent some of the most profound emblems of human diversity, embodying unique ways of perceiving, relating to, and interpreting the world. Globally, approximately 7,000 languages are spoken, yet nearly 3000 of these are endangered, placing almost half of the planet's linguistic heritage at risk. A significant proportion of Indigenous languages remain unwritten, but this does not diminish their cultural richness. They thrive through oral traditions, songs, poetry, and rituals that have sustained communities across generations.

Unfortunately, the loss of indigenous languages is a profound issue. The younger generation often lacks fluency in their ancestral languages due to the dominance of global languages such as English. Without intervention, the ongoing erosion of Indigenous languages threatens to irreversibly diminish our collective cultural knowledge and heritage.

One major factor contributing to this decline is job relocation and urbanization, as individuals frequently move to different countries in search of better opportunities. Technological adoption and digital communication, particularly through social media, further promote widely spoken languages, consequently marginalizing those less common. Preservation of indigenous languages is essential for safeguarding the cultural integrity and dignity of indigenous populations, as Professor Megan Davis of the Permanent Forum on Indigenous Issues highlights. By using technology as a tool rather than a replacement, this project offers a modern way to keep the community's language alive.

1.2 CHOSEN SUSTAINABLE DEVELOPMENT GOAL

The Language Town focuses on **SDG 4: Quality Education**, which aims to ensure inclusive, equitable quality education and promote lifelong learning opportunities. Many indigenous languages are facing risks of disappearing because traditional learning methods are boring, unavailable and difficult for young generations. Language Town will provide interactive and attractive platforms for young generations to learn languages while playing games. This makes education engaging, accessible, and helps preserve languages for the future.

It also supports **SDG 10: Reduced Inequalities** by empowering marginalized linguistic communities and **SDG 11: Sustainable Cities and Communities** in safeguarding intangible cultural heritage.

1.3 PROBLEM STATEMENT

A world of culture is lost every time an indigenous language falls silent due to the lack of accessible learning resources. Many indigenous languages do not have standardized textbooks, storybooks, or grammar guides. Unlike global languages (English, Mandarin, Spanish), they often lack learning apps, online dictionaries, or translation platforms. Elder speakers may not have teaching training or materials. Without these resources, young people have no structured way to learn, even if they want to.

Besides, another reason why indigenous language falls silent due to generational gap in engagement. Younger generations are more exposed to dominant languages through school, media, and technology. Since we live in a globalised world, dominant languages act as "gatekeepers" to opportunity. Youth naturally prioritize proficiency in that language to secure a stable future. Indigenous languages may be seen as less useful for careers or global communication, reducing motivation to learn. However, traditional preservation efforts usually focus on static archiving rather than active use thus it fails to attract young generation.

1.4 OBJECTIVES

a. To spark public interest in learning indigenous languages through gamification
b. To preserve indigenous languages using AI and revitalizing their everyday use

2.0 REVIEW OF EXISTING SOLUTIONS

The global challenge of Indigenous language loss has prompted numerous preservation efforts, yet most people do not integrate these languages into their daily life. Organizations such as the United Nations Educational, Scientific and Cultural Organization (UNESCO) focus on creating dictionaries, grammar guides, and building rich multimedia archives. However, these initiatives often treat indigenous languages as objects of study rather than as living means of communication, which make it easier for some people to learn their language.

Several applications that are available on the market, for example Gurray, IndyLan and FirstVoices that teach indigenous language through games and rewards designed to attract users. Although these applications provide engaging learning experiences, they are often modeled on global language learning frameworks and lack integration of cultural traditions. Social media platforms also host content related to indigenous languages, particularly on YouTube, where individuals share lesson and discussions. Yet, this content is typically unstructured, lacks contextual relevance, and does little to strengthen users' practical fluency.

Despite the existence of organizations, applications and platforms, there are several gaps and limitations that hinder Indigenous languages from thriving in everyday life. Firstly, dictionaries, grammar guides and multimedia archives keep indigenous languages as static historical artifacts rather than living tools of communication, failing to inspire younger generations. Secondly, language-learning applications often emphasize vocabulary and grammar but neglect real-life oral practices, rendering the learning process less meaningful. Finally, while social media provides visibility, Indigenous languages content remains fragmented, unorganized, and overshadowed by dominant global languages.

Therefore, our project aims to fix these problems by combining game-based learning with AI-supported tools. Through interactive challenges, rewards, and competitive activities, Indigenous language and AI supported learning which transforms indigenous language learning into learning enjoyable and motivating through game-based challenges, rewards, and competitive activities, which encourage users to practice regularly. At the same time, AI tools give personalized feedback, helping learners improve their pronunciation and skills.

3.0 METHODOLOGY

There are 3 phases which are included in our design thinking process:

Phase 1: Empathize

Indigenous language is crucial for preserving cultural and linguistic variety and plays a significant role in shaping one's identity. Malaysia is a diverse nation of several races and ethnic groups. The major ethnic groups of Malaysia are the Malays, Chinese and Indians, each with their own languages and dialects. There are 18 distinct minority groups of Orang Asli tribes in Peninsular Malaysia, including Temiar, Jakun, Semang, and Mah Meri. The pervasive use of English as a global language in Malaysia poses significant challenges to the preservation of numerous indigenous languages. Malaysia has over 130 indigenous languages, including languages like Iban and Kadazandusun in Borneo, and Orang Asli languages in Peninsular Malaysia. As part of this study, I conducted interviews with Dayak friends representing different age groups in Sarawak. Recent data indicate a language shift in the Dayak demographic in Sarawak. Seniors over 60 persistently use Kelabit in their daily discourse, while middle-aged adults aged 30-60 maintain a dual language competency, engaging in Dayak alongside English or Kelabit. In contrast, the younger generation speaks almost exclusively in dominant languages, particularly English. This decline in Indigenous language use stems largely from the second generation's decision to stop speaking Kelabit at home, which disrupted its transmission to younger family members.

Phase 2: Define

The interviews with the Dayak community in Sarawak reveal that while elders continue to speak Kelabit fluently, younger generations are increasingly shifting toward dominant languages. This pattern reflects significant absence of accessible learning resources for Indigenous languages. Unlike global languages such as English, Mandarin, or Spanish, Kelabit lacks standardized textbooks, grammar guides, storybooks, or digital learning platforms. Elder speakers often rely on oral transmission, but they are not equipped with formal teaching materials or educational training. Consequently, even when younger individuals express interest in learning their ancestral language, they have no structured pathway to acquire it.

The generational differences observed in the interviews further highlight the breakdown of intergenerational transmission. Elders maintain strong cultural attachment to Kelabit, but the second generation's bilingualism and reduced use of the language at home have disrupted continuity. In a globalized society, language barriers erected by dominant languages lead younger generations to prefer English or Malay for educational and career advancement. Consequently, their perceived relevance, particularly in terms of daily life, is lost, causing a disconnection with indigenous languages such as Kelabit. Preservation efforts have largely focused on documentation, but passive measures alone are insufficient. Without proactive strategies like language integration in education and community practices, indigenous languages face continuous decline.

Phase 3: Ideate

We have decided to introduce “Language Town” in this project as an initiative to preserve indigenous languages in Malaysia. “Language Town” is structured as a series of step-by-step lessons, each designed to be completed before progressing to the next chapter. Every lesson guides learners in acquiring simple vocabulary, which is presented systematically in individual boxes accompanied by pronunciation. This format allows learners to read and understand the vocabulary with ease. Upon completing the vocabulary section, learners are then directed to a small quiz for deeper understanding. Users will be rewarded with coins after completing one lesson. These coins can be used to purchase accessories and clothing for their in-game character. This feature allows younger learners to personalize and decorate their character according to their own preferences.

Additionally, “Language Town” features an AI-driven radio module that generates both audio and transcriptions. Learners can sharpen their listening comprehension through randomly generated AI broadcasts. Each Kelabit transcript is translated into English, allowing learners to grasp the original meaning while exploring the language's unique speech patterns and grammatical structures. Beyond full sentences, learners can also input individual vocabulary words into a designated text box for instant English translation.

Furthermore, one of the distinctive features of “Language Town” is that users can design and develop their own town. At the beginning, the game character is unable to speak. After completing the first lesson, the character begins to communicate using the vocabulary acquired in that lesson. As learners progress through subsequent lessons, additional vocabulary is unlocked. New game characters also become available, allowing them to interact with one another in everyday scenarios. This feature facilitates the learning of indigenous languages by embedding vocabulary and grammar practice within context-based, in-game texts and situations.

4.0 RESULTS

This project aimed to spark public interest in learning indigenous languages through gamification and to preserve languages using artificial intelligence (AI), thereby revitalizing their everyday use. Based on the development and testing of the Language Town prototype designed in Figma, the system demonstrated that a gamified learning environment combined with AI-assisted tools can significantly enhance user engagement and accessibility in Indigenous language learning.

Our system integrates game elements such as missions, rewards and level progression to make the learning process of indigenous languages more interesting and engaging. Users must complete all tasks or challenges to unlock new vocabulary and phrases, which enable communication with the villagers within the system. Upon completing tasks, learners receive virtual coins that can be used in the game shop to customize their characters. This approach creates a more interactive learning experience compared to traditional methods which only learn by completing all levels without meaningful interaction.

The systems also incorporate interactions with the non-player characters (NPCs), represented by villagers who guide and teach users after they successfully unlock new levels. Initially, villagers will remain silent when users enter the game. As tasks are completed and vocabulary is acquired, villagers begin to speak and interact with learners. This stimulates real-life communication scenarios, allowing learners to apply newly learned vocabulary rather than simply memorizing it. This helps users to reinforce language retention and build confidence in using Indigenous language.

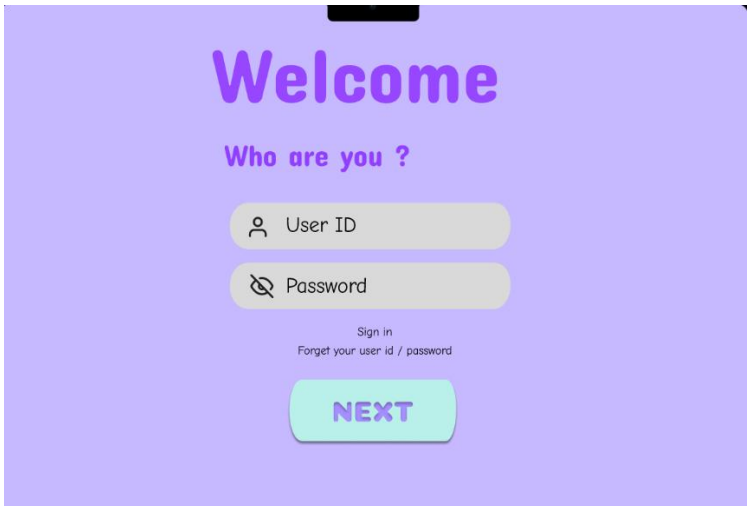
In addition to gamification features, the system integrates artificial intelligence (AI) to enhance the learning experience. The AI module generates spoken language content alongside text translation which exposes learners to sentence structures, pronunciation patterns, and listening comprehension practice. Users will learn and understand the meaning and

pronunciation of the language more effectively. Users can input individual words or sentences and receive translations. This will reduce the difficult understanding of new language elements.

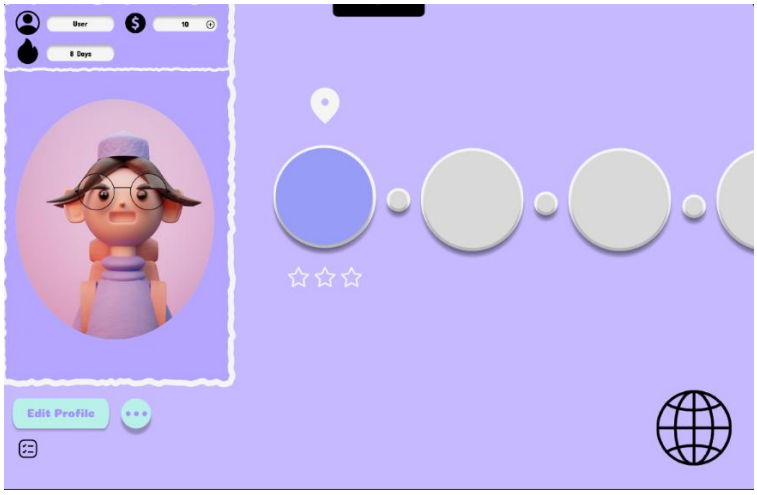
From the prototype evaluation, the interface design was easy to navigate and friendly for every user. Key interfaces such as login page, home dashboard, lesson interface and mission panels were successfully designed to guide users through a clear leaning pathway. The reward system and mission panels further help users to monitor their achievements and maintain consistent learning habits.

Overall, the results demonstrate that the prototype successfully meets the objectives of the project. The integration of gamification and AI features creates an engaging digital environment that motivates learners to explore Indigenous language. By combining structured lessons, interactions with NPCs and AI-assisted learning tools, our project provides a modern approach to encourage younger generations to learn indigenous language.

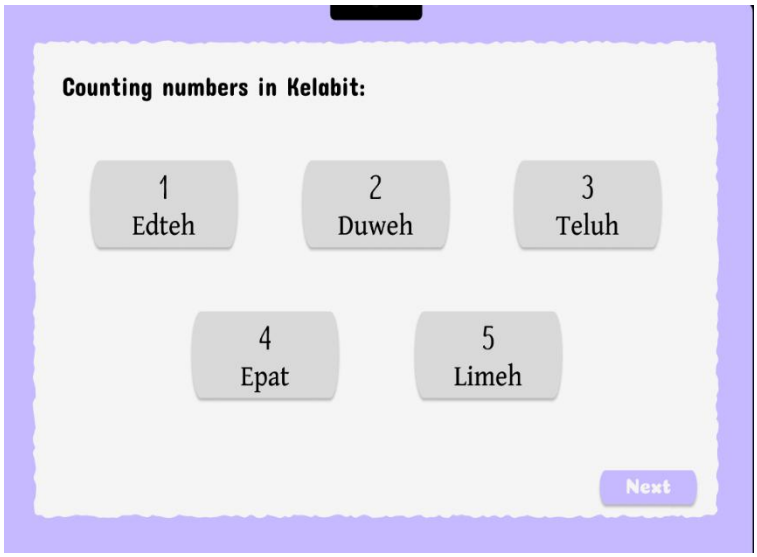
5.0 PROTOTYPE SHOWCASE



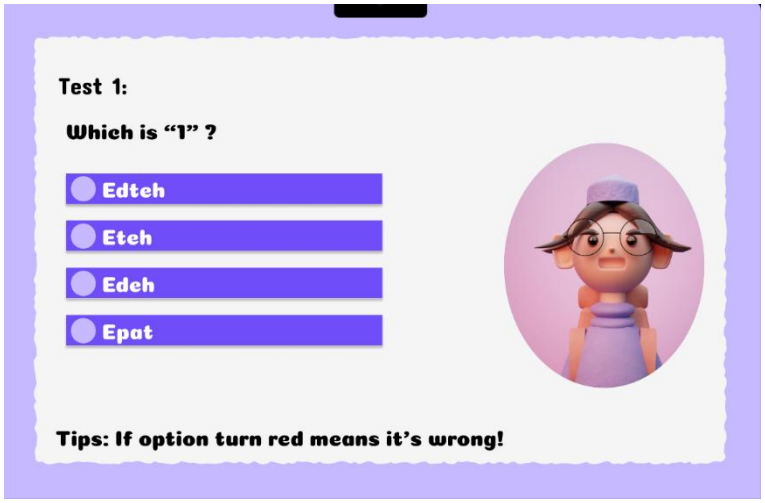
Login page: provides secure access and personalized user identification before entering "Language Town".



Home Dashboard: progress tracker with rewards (streaks, star awards), profile editing, and global access.



Lesson screen: introduces Kelabit numbers in a clear, step-by-step format for easy learning.

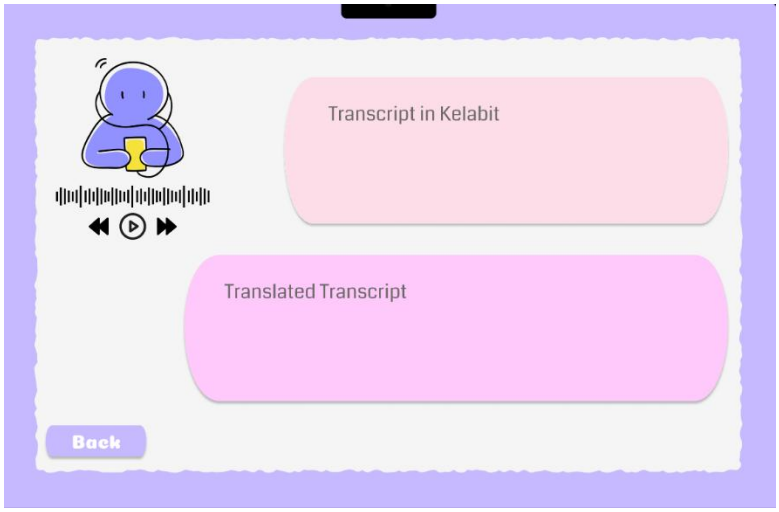


Interactive test: strengthens memory by checking answers and guiding correction.

Reward screen:

congratulates learners and motivates them with coin rewards.

Translation module: converts inserted text into Kelabit instantly, like Google Translate



AI radio module converts spoken language into transcripts with real-time translation support

Virtual shop: allows learners to decorate their character using coins earned from lessons.

Mission panel: rewards progress with coins, encouraging consistent learning.

Small town module: villagers begin silent, but completing lessons to unlocks vocabulary and enable them to speak

Progressive learning: villagers start speaking once learners master and unlock new words.

6.0 AI ACKNOWLEDGEMENT

The development of this project utilized artificial intelligence (AI) tools to enhance efficiency, creativity, and linguistic precision. We fully disclose their use to ensure transparency.

Specifically, **Google Gemini** was employed for initial translation tasks and to assist in refined word selection. Furthermore, **ChatHub** was used as a platform to access and compare outputs from multiple AI models—including **ChatGPT**, **DeepSeek**, and **Gemini**—simultaneously. This multi-model approach was instrumental in enriching ideas, refining translations, and selecting the most effective language for project documentation and conceptual development.

By integrating these advanced technologies, we were able to bridge the gap between ancient heritage and modern innovation, ensuring that the preservation of indigenous languages is both technically robust and culturally resonant. Without these tools, the depth and efficiency of this revitalization effort would not have been possible. We acknowledge that these tools served as collaborative assistance, helping to streamline the creative and analytical process, while all final decisions, integrations, and project directions remain in the work of the human team.

7.0 CONCLUSION

Our project “Language Town” demonstrates a clear, actionable pathway for revitalizing indigenous languages by combining the motivating power of gamification with the scalability and adaptability of AI. By wrapping vocabulary learning, contextual practice and pronunciation exposure into narrative quests, quiz-based interactions and an AI-driven audio radio, the project turns language preservation from a passive documentation effort into an engaging, repeatable habit for learners, especially youth who are most likely to shape future language use. The project does more than teach words; it rebuilds everyday use by embedding new vocabulary and grammar into simulated social contexts where learners can practice, personalize and see progress through in-game rewards and town development. Our solution directly tackles the root causes of language decline the lack of accessible, engaging learning resources for younger generations and the breakdown of intergenerational transmission. By transforming language acquisition into an immersive, rewarding game, we aim to rebuild the perceived relevance and daily utility of languages like Kelabit, moving preservation from a passive archival effort into an active, community-driven revival.

Social and cultural impact is central to the project’s value. The project “Language Town” targets not only individual learners but also communities and educators, offering tools that can reduce inequalities in language access and support cultural continuity. By aligning with SDG 4 Quality Education, SDG 10 Reduced Inequalities and SDG 11 Sustainable Cities and Communities, the initiative positions indigenous languages as living assets that contribute to identity, intergenerational connection and cultural resilience. When elders, families and young people can share the same playful platform, intergenerational transmission becomes more likely and more natural.

AI is a pivotal enabler, but it must be applied thoughtfully. Automated translation, synthetic voiceovers and dynamic content generation allow rapid expansion across languages and dialects, also provide scalable listening and speaking practice where pedagogical resources are scarce. Yet the AI components should be continuously validated and tuned with community input so that translations, pronunciations and cultural contexts remain accurate and respectful.

In doing so, the project avoids flattening linguistic nuances and ensures technology amplifies, rather than replacing native expertise.

To scale impact while maintaining quality, robust evaluation and evidence collection are essential. Define clear metrics, vocabulary retention, listening comprehension gains, frequency of in-community use, number of community-contributed assets, and also long-term retention across age groups and embed tools to measure them. Run pilot studies with pre-post assessments, qualitative interviews with elders and learners, and analytics for in-game progression. Use these findings to refine pedagogy, tune AI models and demonstrate the program's effectiveness to funders and cultural institutions.

Ethical, cultural and accessibility considerations should guide every technical decision. Ensure consent and benefit-sharing for community recordings, provide options for privacy, ownership rights and offline access where internet connectivity is limited. Design interfaces and reward systems that avoid cultural commodification, prioritize cultural sensitivity reviews, retain glossaries of cultural terms and include elders or cultural committees in approval of workflows. Accessibility features such as larger text, visual cues and low-bandwidth modes will broaden reach to families and educators with diverse needs.

Practical future improvements (actionable next steps):

- Co-creation toolkit: build simple recording, editing and annotation tools so communities can contribute validated vocabulary, stories and pronunciations with minimal technical barriers.
- Human-in-the-loop AI: establish workflows where AI outputs such as translations, voice synthesis is reviewed and corrected by native speakers and linguists to ensure high fidelity and cultural accuracy.
- Curriculum alignment: partner with educators to align lessons with local school syllabi or community teaching programs, enabling use in classroom and family settings.
- Adaptive learning engine: implement spaced repetition and adaptive difficulty, so the system personalizes practice and maximizes long-term retention for each learner.

- Community hubs and multiplayer features: add cooperative town events, family accounts and teacher dashboards to foster shared practice and classroom integration.
- Offline and low-bandwidth modes: create downloadable lesson packs and lightweight audio so remote or low-connectivity communities can use the app without constant internet access.
- Impact evaluation plan: run randomized pilots or controlled rollouts with measurable learning outcomes and sociolinguistic indicators to validate effectiveness and inform scale-up.
- Cultural safeguards: formalize consent, copyright and benefit-sharing policies, so communities retain ownership and receive recognition or compensation for their content.
- Expand language coverage responsibly: prioritize languages based on community desire and capacity to participate and use modular tools to accelerate safe, community-validated onboarding of new languages.

In sum, the project "Language Town" is a compelling, scalable approach to transforming how indigenous languages are learned and lived. With thoughtful co-design, rigorous evaluation, ethical safeguards and targeted technical enhancements, the project can move from a promising prototype to a sustainable platform that restores everyday use of endangered languages and strengthens cultural continuity for generations to come.